

## GROWRIDER · DESIGN DECISION RECORD

# GrowRider recommendations accepted

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## DOCUMENT CONTROL

|             |                                                               |         |                |
|-------------|---------------------------------------------------------------|---------|----------------|
| Document ID | GRR-DDR-002                                                   | Version | 0.2            |
| Status      | <b>Draft</b>                                                  | Date    | 2026-09-26     |
| Author      | Amish Chadha                                                  | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept                                 |         |                |
| Repository  | github.com/BoujeeEnjinia1701/growrider                        |         |                |
| Source      | docs/decisions/0002-recommendations-accepted.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                       |
|---------|------------|--------------|----------------------------------------------|
| 0.2     | 2026-09-26 | Amish Chadha | Budget top-up to \$300 decided by Amish (N1) |
| 0.1     | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002)  |

## 0002: Recommendations accepted

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- **Date:** 2026-09-25
- **Status:** accepted for items D7 to D13; D1 to D6 confirmed; items O1 to O4 and N1 to N2 remain proposed

### ■ Context

The TRL 3 review note ( docs/REVIEW.md , session 2026-09-25: TRL 3) listed seven new items as "Proposed, awaiting Amish", each with a recommendation, and kept four TRL 2 items open because no recommendation had been made. GRR-DDR-001 had recorded D1 to D6 as "Adopted as recommended" under Amish's instruction of the same day. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item with a recommendation is therefore decided in favor of that recommendation; where several options were offered, the recommended option is the decision. Items without a recommendation stay open. TRL 4 remains on hold by Amish's instruction, so any decided item that needs building, testing or purchasing is recorded as decided but on hold.

### ■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-25: TRL 3) and in GRR-CAL-001 v0.1.

### ■ Decision

*Table 1. Items decided by this record.*

| #              | Item                                                                                  | Decision                                                                                                                                                                                                   | What changed in the repo                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D1<br>to<br>D6 | Wheel size, brakes, frame, tires, crank length, \$120 production target (GRR-DDR-001) | Decided by Amish, 2026-09-25: go with recommendation (confirmed)                                                                                                                                           | Wording in GRR-DDR-001 and REVIEW unchanged in substance                                                                                                                                                                                                                                                                                                                                                                                                           |
| D7             | Mass (R5)                                                                             | Decided by Amish, 2026-09-25: go with recommendation. Option (b): chromoly main tubes, an aluminium rack, alloy rims and an alloy bar, keeping solid tires (D4); R5 stays at 13 kg as the production goal  | cad/src/model.py : seat tube 31.8 x 1.2 mm, down tube 31.8 x 0.9 mm and loop tube 28.6 x 0.9 mm in chromoly 4130; rack 12 x 1.5 mm aluminium 6061-T6. BOM lines 1, 6, 8 and 13 respecified and repriced. GRR-CAL-001 v0.2: mass 15.3 kg to 14.4 kg (the TRL 3 note estimated 13.6 kg; the model-based figure is higher because the seat tube keeps a 1.2 mm wall for the sleeve and the head tube keeps 1.5 mm). R5 restated as the production goal; still not met |
| D8             | Alloy front rim (R7)                                                                  | Decided by Amish, 2026-09-25: go with recommendation. Alloy front rim specified                                                                                                                            | BOM line 7 now alloy only; R7 target names the alloy rim; wet front braking for the smallest rider about 0.15 g instead of 0.05 g with steel                                                                                                                                                                                                                                                                                                                       |
| D9             | Coaster hub brake data (R7)                                                           | Decided by Amish, 2026-09-25: go with recommendation. Obtain the brake ratio of a regional coaster hub before further brake work                                                                           | Recorded as an open action; the data must come from a hub maker or the co-design partner. Any hub purchase or bench test is TRL 4 work, on hold                                                                                                                                                                                                                                                                                                                    |
| D10            | Strength (R12)                                                                        | Decided by Amish, 2026-09-25: go with recommendation. Chromoly steerer and quill, and a 1.8 mm wall sleeve; recheck in GRR-CAL-001                                                                         | Sleeve 28.6 x 1.5 mm became 29.2 x 1.8 mm chromoly (25.6 mm bore for the 25.4 mm post), which needs the 29.4 mm bore of a 1.2 mm seat tube. Steerer and quill chromoly at the same size. GRR-CAL-001 v0.2 adds a 90 MPa screen for chromoly sections: 3 of 6 sections above the screen became 1 of 6 (steerer at the crown, 117 MPa). Drawing GRR-DWG-001 to Rev P2                                                                                                |
| D11            | R2 target                                                                             | Decided by Amish, 2026-09-25: go with recommendation. R2 widened from 436 to 654 mm to 400 to 670 mm                                                                                                       | GRR-REQ-001 v0.4; the design already meets it                                                                                                                                                                                                                                                                                                                                                                                                                      |
| D12            | R9 tools                                                                              | Decided by Amish, 2026-09-25: go with recommendation. R9 split into a parent tool list (13 mm spanner, screwdriver) and a mechanic tool list (13, 15 and 32 mm spanners, BB lockring spanner, screwdriver) | GRR-REQ-001 v0.4 (R9a, R9b); both tool lists met on paper; the 10 min fit change sits at the limit, so R9 stays at risk                                                                                                                                                                                                                                                                                                                                            |
| D13            | Low bar option                                                                        | Decided by Amish, 2026-09-25: go with recommendation. A flat bar for the smallest riders is a co-design question                                                                                           | Added to the open questions of GRR-PRB-001 v0.4 and GRR-PRC-001 v0.4; no design change                                                                                                                                                                                                                                                                                                                                                                             |

Table 2. Items still open.

| #  | Item                                                                                                                        | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| O1 | Helmet funding (BOM line 20, \$12)                                                                                          | Proposed, awaiting Amish (no recommendation was made). Since the 2026-09-26 top-up (N1), the \$293 total with the helmet fits the \$300 budget                                                                                                                                                                                                                                                                                                                                                               |
| O2 | First partner                                                                                                               | Proposed, awaiting Amish; community designs pick co-design partners per area later                                                                                                                                                                                                                                                                                                                                                                                                                           |
| O3 | First region (Zambia, Kenya or Malawi suggested)                                                                            | Proposed, awaiting Amish (no recommendation was made)                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| O4 | Co-design partner for fit measurements and the regional parts list                                                          | Open; picked per area later                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| N1 | Prototype budget: D7 and D10 raise the bike to \$281 (\$293 with the helmet) against the \$250 in <code>project.yaml</code> | New, proposed, awaiting Amish. Options: raise the budget to \$300; keep \$250 and build the first prototype in hi-tensile steel with the chromoly parts later; or keep \$250 and treat R11 as not met. Recommendation: raise to \$300, which also settles O1. Budget top-up to \$300: decided by Amish, 2026-09-26. <code>budget_usd</code> is now \$300; GRR-REQ-001 v0.5 and GRR-CAL-001 v0.3 record R11 as at risk (prototype \$281, \$293 with the helmet, within budget; production cost not estimated) |
| N2 | Steerer at the crown still above its screen (117 MPa against 90 MPa; 98 MPa with a 25.4 x 2.0 mm steerer)                   | New, proposed, awaiting Amish. Recommendation: specify a 25.4 x 2.0 mm butted chromoly steerer and confirm against the ISO 4210-2 fork tests at TRL 4                                                                                                                                                                                                                                                                                                                                                        |

## ■ Consequences

- GRR-REQ-001, GRR-PRC-001 and GRR-PRB-001 go to v0.4 and GRR-CAL-001 to v0.2; drawing GRR-DWG-001 goes to Rev P2; the STEP, STL and media files are regenerated from the model.
- Requirement status after this record (GRR-CAL-001 v0.2): 5 met, 2 not met (R5 mass 14.4 kg; R11 prototype cost \$281), 3 at risk (R7, R9, R12), 2 not verifiable at TRL 3. See GRR-CAL-001, Table 6.
- `trl` and `trl_target` stay at 3. TRL 4 is on hold by Amish's instruction; nothing in this record authorizes building, testing or purchasing.